Two-Grammar Decomposition of Interaction in an Anonymous Two-Channel Closed Wave System

A Classification Theorem for Conserved Readouts, Term-by-Term Machine-Precision Verification of the Two-Term Flow Identity, and the Necessary Bifurcation of Gravity-Type and Charge-Type Composition Laws from Wave Shape

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Abstract

(Discovery of the mechanism) The two-body flow in an anonymous closed wave system decomposes identically into

$$dN_B \;=\; \underbrace{\sin^2\theta\,(N_A - N_B)}_{\text{magnitude term}} \;+\; \underbrace{\sin 2\theta\,\mathrm{Re}\langle a|b\rangle}_{\text{overlap term}}$$

two terms, and no third term exists. These two terms read, respectively and **exclusively**, the two kinds of memory carried by the shape of a wave — the norm (magnitude), and the relative phase on shared modes (shape). Consequently, in a wave system possessing no labels, no background, and no external fields, interaction necessarily differentiates into exactly **two kinds**:

	Magnitude term	Overlap term
Composition law	**Additive** (mean, error $\leq 7.2 \times 10^{-16}$)	**Product** $\sqrt{f_A f_B}$ (error $\leq 4.0 \times 10^{-15}$)
Sign	None (only the norm difference sets the direction)	Present (reverses with relative phase, measured)
Screening	Impossible (always on whenever a norm is present)	Possible (two mechanisms: zero component / non-shared channels)
Quantization	None (continuous)	Present (rational locking, measured)
Range of the coupling	Ratio $(R/R_0)^2$: unbounded, continuous (**weakness by orders of magnitude is possible**)	Angle: unitarity bound $\alpha \leq 1/4\pi$ $\times$ lock discretization (**confined to the $O(10^{-1})$ band**)

This property table **corresponds one-to-one with the contrast table of gravity (additive mass, no sign, unscreenable, unquantized) and electromagnetism (product of charges, $\pm$, screenable, quantized)**. This correspondence is not an arbitrary assignment based on resemblance of properties: if the overlap term is read as gravity-type, gravitational action would reverse and vanish under relative-phase reversal and channel non-sharing, contradicting universality; if the magnitude term is read as charge-type, none of the amplitude product, the two signs, or neutralization can be constructed — **the assignment is a unique correspondence that excludes the reversed assignment** (Section 6.5, Corollary). That is, this paper discovers the mechanism by which the divergence in the properties of the two great long-range interactions arises necessarily, from within a single identity, as the exclusive readout of the two memories

of wave shape, and verifies every row at machine precision. Anonymity is never broken — distinguishing the forces requires neither particle species nor external input of coupling constants.

(Completeness on the Coulomb side) The overlap term possesses all five elements of the Coulomb structure — inverse square (already derived from harmonic closure [3]), product of couplings, sign, quantization, and value (the exact finite-order root $\sin^2(23\pi/124)$, matching the dimensionless elementary charge $\sqrt{4\pi\alpha}$ to 0.16 ppm [2]). **We call this quantity the model's elementary charge.**

(Hierarchy) A long-range interaction that is weaker by orders of magnitude can exist only in the grammar of the magnitude term. Moreover, the coupling of the magnitude term is of scale-ratio type $(R/R_0)^2$, and in any universe containing localized closures $R \ll R_0$ — **the enormous hierarchy between the two great long-range forces is not a coincidence requiring explanation but a structural necessity that follows from the very definition of localization: particles are far smaller than the background**. What remains underived is not the existence of the hierarchy but only its exponent (the scale dictionary).

(The one remaining item) Toward identification with the physical elementary charge, the only work remaining **inside the model** is the derivation of the mechanism that selects order 124. The static decomposition does not select it (refuted); the fixed-point channel has been eliminated; the lock weights collapse with the denominator (measured ratio > 461); and observational selection passes the selection down to the divisor structure of the observation clock — the selection problem has been transformed into a single falsifiable question, "why is the observation clock commensurable with 248?" (sister paper C).

(Convergence) The force law product × cos(relative phase) × inverse square was realized in fluids by Bjerknes (1875, pulsating spheres); mode overlap as the existence condition of coupling is realized by the visibility law of optics; quantization plateaus via locking by the Shapiro steps of superconductivity; and the correspondence between force ratios and scale ratios by Dirac's large numbers (1937) — each independently realized and observed. That mutually unrelated systems converge on the same grammar is evidence that this grammar is not a detail of the medium but a general structure of closed wave systems. The contribution specific to this paper is to show that, in an endogenous-coupling system in which the coupling is not supplied externally, all five elements and every row of the two grammars are derived and verified simultaneously as terms of a single flow equation, and survive an audit over five variants of the readout convention.

Keywords: two-channel exchange scattering, conserved readouts, composition laws, interference cross term, Bjerknes force, Arnold tongue, charge quantization, hierarchy problem, reproducible computation

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0. Conclusion

> The two-body flow of an anonymous closed wave system decomposes identically into a "magnitude term" and an "overla
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> Magnitude term: additive composition (error 7.2×10^{-16}), no sign, unscreenable; its readout is exactly conserved.
>
> Overlap term: product composition $\sqrt{f_A f_B}$ (error 4.0×10^{-15}), sign = relative phase, neutralized by two mechanisms,
>
> This contrast corresponds one-to-one with the property table of gravity/electromagnetism, and only the magnitude terr

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Position of This Paper and the Three-Part Structure

This work reports results obtained from a single experimental environment, divided into three parts according to the independence of the claims.

	Subject	Main results	Status
Part I (this paper)	Two-grammar decomposition of interaction	Two-term flow identity, classification theorem for conserved readouts, composition laws (additive/product), sign, neutralization, direction of the hierarchy	This paper
Part II (sister paper B)	Structure of counting readouts	Exact rationality of equal-weight points, meta counting rule (bin table $\times$ mask), full report of the convention audit, complete verification of the Niven intersections (orders 3/4/6 = crystallographic menu), readout device for the integer pair (m, n)	Published (DOI: 10.5281/zenodo.21763998)
Part III (sister paper C)	Lock dynamics and the limits of selection	Full report of the order-6 Arnold tongue, measured tongue widths (ratio > 461), fixed-point map (elimination of the selection channel), observational selection (clock inheritance and finite-observation limits), supercritical diagnostics	Published (DOI: 10.5281/zenodo.21764000)

The interdependence is as follows. Claims 1–3 of this paper stand on their own; the results of sister papers B and C are cited only as a concrete instance of the quantization row (summarized in Section 7) and as the guarantee of convention robustness (summarized in Section 8). Sister paper B shows in complete form that the claims of this paper do not depend on the readout convention. Sister paper C carries the search program for Claim 4 (the selection problem). All three parts share the reproduction package of the same repository (Section 12).

1. Research Question

In a system consisting only of waves without labels, how does interaction differentiate into kinds? Without externally supplying particle species, coupling constants, or background fields, can forces of the distinct grammars — gravity-type (universal, additive, unsigned) and charge-type (product, both signs, quantized) — arise simultaneously from the same substrate? If they can, which information in the wave shape carries the distinction, and can the exhaustiveness of the distinction (the absence of a third type) be established?

In prior work, the operator structure and finite-order resonance of the two-channel exchange-scattering system [2], the inverse-square law from harmonic closure [3], and conserved-quantity readout via interference [4] were each established separately. This paper unifies them on a single flow equation and takes the mechanism of differentiation itself as the main result.

2. Definition of the System and Design Boundary

The system is the two-channel complex field $a(\chi, \eta), b(\chi, \eta)$ of the prior experimental environment (the paper-9-series control-experiment / wave-packet-contraction execution environment; repository directory romanized as `taishou-jikken_hasoku-shuushuku_jikkou-kankyou_v1/ab_invariant_theta_toy_v1`).

States use only the harmonic-packet construction via the existing `explicit_packet_case` / `make_case_state`, and in the standard configuration a carrier ($q_A = +1, q_B = -1$) is attached. The readout θ is defined from the power ratio of the even, $|k| \geq 4$ sector of the joint χ spectrum as

$$\theta = \operatorname{atan2}\Big(\sqrt{P_f},\, \sqrt{P_b}\Big), \qquad P_f = \sum_{\text{mask}} \left(|A_k|^2 + |B_k|^2\right)$$

and one collision is the real orthogonal rotation

$$a' = a\cos\theta - b\sin\theta, \qquad b' = a\sin\theta + b\cos\theta$$

(the sign convention of the rotation was fixed to $\sigma = +1$ by a calibration experiment).

Design boundary: The scattering body and the θ-readout formula are unchanged across all experiments. Target values (fractions f, etc.) are used only for state construction and are never passed to the forward processing. The variant readouts and probe rotations of Section 6 are distinguished as explicitly declared working hypotheses / instrument settings.

Intrinsic transmission fraction: The fraction $f(s)$ of a state s is defined as the in-mask power ratio of s alone. Via the orthogonal two-harmonic mixture

$$s(f) = \sqrt{1-f}\, s_{\text{bos}} + \sqrt{f}\, s_{\text{fer}}$$

of the all-bosonic single harmonic ($f = 0$) and the all-fermionic single harmonic ($f = 1$), f can be tuned continuously and exactly (fraction error $\leq 10^{-31}$, cross overlap $\leq 1.3 \times 10^{-16}$).

3. The Two-Term Flow Identity

Direct computation for the real orthogonal rotation gives the per-collision change of the B-channel norm as

$$\boxed{dN_B = \underbrace{\sin^2\theta\,(N_A - N_B)}_{\text{magnitude term}} + \underbrace{\sin 2\theta\, \operatorname{Re}\langle a|b\rangle}_{\text{overlap term}}}$$

This is an identity of the norm algebra of rotations, and **no third term exists in this class of two-channel unitary flows**. The magnitude term reads only norms (diagonal quantities); the overlap term reads only the inner product $\langle a|b\rangle$ (the amplitude product and relative phase on shared modes). All experiments below measure, term by term, the composition law, sign, conservation, and quantization of these two terms.

Note that the vanishing of the magnitude term at $N_A = N_B$ does not mean the vanishing of the coupling; it means that the net transfer flow balances between two symmetric channels. The coupling coefficient $\sin^2\theta$ and the observed net flow $\sin^2\theta\,(N_A - N_B)$ are distinct — the same distinction as the fact that thermal coupling between two bodies at equal temperature does not vanish even though the net heat flow is zero.

For the mixed states $a = s_A(f_A)$, $b = \nu\,[\sqrt{1-f_B}\, B_1 + e^{i\varphi}\sqrt{f_B}\, B_7]$ (ν: norm multiplier), the exact prediction is

$$dN_B = \sin^2\theta\,(1-\nu^2) + \sin 2\theta\, \nu \Big[\sqrt{(1-f_A)(1-f_B)}\, c_0 + \sqrt{f_A f_B}\, c_6 \cos\varphi\Big]$$

where $c_0 = \langle A_1|B_1\rangle$ and $c_6 = \langle A_5|B_7\rangle$ are couplings measured from the state construction alone.

4. Classification Theorem for Conserved Readouts

Theorem (sufficient condition for conservation). The real orthogonal rotation exactly conserves $|A_k|^2 + |B_k|^2$ in each frequency bin (the parallelogram identity). Hence any readout that is a function only of the bin-wise rotation invariants is exactly conserved under the evolution.

Numerical classification. Seven candidate θ readouts were classified by their self-drift under the dynamics that each readout itself drives (300 collisions $\times$ 3 amplitudes).

Readout	Classification	Max self-drift
R1 diagonal sum $\sum(\|A_k\|^2 + \|B_k\|^2)$	**Conserved (unique)**	2.2×10^{-16}
R2 $\|A + B\|^2$ (X power)	Dynamical	1.06
R3 $\|A - B\|^2$	Dynamical	1.06
R4 cross spectrum $\|\sum A_k B_k^*\|$	Dynamical (fixed-point convergence)	0.43
R5 $\|A\|^2$ only / R6 $\|B\|^2$ only	Dynamical	1.35
R7 cross real part	Dynamical	0.38

Only the diagonal sum was conserved. This single structure simultaneously generates the three no-gos described below — the freezing of θ, additive composition, and the absence of product structure in the diagonal sector. Note that R4 was initially misidentified as conserved on the basis of its tail standard deviation of 10^{-16}, but the classification experiment revealed convergence to a fixed-point attractor (the history of the correction is recorded in the Reproducibility section).

5. Experiment I: Composition Law of the Magnitude Term (Additive, Gravity Grammar)

The effective exchange strength $R_{\text{joint}} = \sin^2\theta$ was measured on a 6×6 grid of intrinsic fractions (f_A, f_B) and compared with three composition-law hypotheses.

Hypothesis	Max error	Verdict
Mean $(f_A + f_B)/2$	7.2×10^{-16}	**Exact agreement**
Product $f_A f_B$	0.5	Rejected
$\sqrt{f_A f_B}$	0.5	Rejected

The coupling of the diagonal sector composes exactly additively (as the mean). This is the grammar corresponding to the additivity of mass, and no vertex product of $q_A q_B$ type exists in this sector (zeroth-order no-go). No product structure was detected in the transfer observables (swings of norm and localization) either (correlation $|r| \leq 0.24$). Furthermore, the mean law does not depend on the readout convention: under the variant masks (even$\geq$6, any$\geq$4) it holds identically for the fractions measured per mask (max error 4.4×10^{-16}; as a theorem, it is a consequence of bin-wise power additivity).

6. Experiment II: Charge Grammar of the Overlap Term (Product, Sign, Neutralization)

6.1 Null Theorem (Carrier Separation = Closure of the Coherent Flow Path)

In the standard configuration (with carrier), over all 7×7 pairs of A/B single harmonics, $|\langle A_j | B_k \rangle| \leq 2.7 \times 10^{-17}$ — the two channels separate into exactly orthogonal subspaces, and the overlap term is identically zero. This is not a failure of measurement but a theorem: it establishes that **the existence condition of charge-type interaction is mode (channel) sharing**, and it provides the second mechanism of neutralization (channel non-sharing).

6.2 Record of the Invalid Experiment

The carrierless + parity-reversed-mask configuration (v2) became invalid: without the carrier, the harmonics do not land on the intended bins and the fraction tuning collapses. This failure established that the diagonal θ readout (which presupposes the carrier) and the coherent flow path (which presupposes sharing) sit on mutually exclusive configurations in this toy model, and that a physical model requires partial sharing (preserved as a record).

6.3 Term-by-Term Verification of the Charge Grammar

With carrierless anchors (shared channels measured: $c_0 = c_6 = 0.500000$, cross-anchor leakage $\leq 5 \times 10^{-16}$) and probe rotation angles $\theta_p \in \{0.05, 0.1, 0.2, 0.3\}$ (declared as instrument settings), we measured the 6×6 grid $\times$ 8 phase points φ $\times$ equal/unequal norm ($N_B = 1.44$). Against the exact prediction of Section 3:

Verification item	Result
Modulation amplitude $M = \lvert\sin 2\theta_p\rvert \sqrt{f_A f_B}\, \nu \, \lvert c_6 \rvert$ (vertex-product law)	Max error 4.0×10^{-15} (equal norm), 5.7×10^{-15} (unequal norm)
Offset (magnitude term + bosonic cross term)	Max error 7.1×10^{-15} (equal), 6.5×10^{-15} (unequal)
Neutrality ($M = 0$ at $f_A f_B = 0$)	$M \leq 4.2 \times 10^{-15}$
Sign (flow direction reverses under a half turn of φ)	Reversal in every cell where the modulation dominates the offset (63 equal / 76 unequal cells)

All three terms of the flow equation — the diffusion term (magnitude-driven), the bosonic cross term, and the charge term (amplitude product $\times$ relative phase) — matched the prediction at machine precision. **The sign is the relative phase on the shared channel, and the magnitude of the coupling is the product of amplitudes.**

As a by-product, the shared channels have a ladder structure: A_k couples only to $B_{k\pm 2}$, and all coupling values are exactly 0.5 (selection rule $\Delta k = \pm 2$).

6.4 Claim-Hardening Experiments (Multiple Channels, Sign Persistence, Partial Sharing, Distance Law)

Four additional experiments were carried out to close the unverified aspects of the claims of the first draft (all runners and data are committed).

(i) Channel additivity of charge. For mixed states carrying two shared channels simultaneously (A_5–B_7 and A_7–B_9), the effective coupling agreed exactly with the **coherent sum of per-channel amplitude products**

$$\langle a|b\rangle = \sum_{\mathrm{ch}} u\, v\, c_{\mathrm{ch}}\, e^{i\varphi_{\mathrm{ch}}}$$

(6×6 two-phase grid, max error 5.7×10^{-15}). Charge adds as amplitude across multiple channels.

(ii) Multi-collision behavior of the sign. Under iteration of a fixed probe rotation, the transfer oscillates in Rabi fashion, and the sign component (the component that reverses under phase reversal) is exactly mirror-symmetric over the entire trajectory (antisymmetry error 2.2×10^{-16}). The sign is therefore exact as the direction of flow at each collision, but **speaking of a net steady flow requires an argument about averaging over the oscillation** — the same situation in which the Bjerknes force is defined as a time average over the acoustic period.

(iii) Partially shared states = simultaneous validity of the two grammars on the same state pair. The exclusivity recorded in Section 6.2 (the diagonal readout requires the carrier; the coherent flow path requires sharing) is resolved by the mixture

$$a = \sqrt{1-w}\, A_{\mathrm{on}} + \sqrt{w}\, A_{\mathrm{off}}, \qquad b = \sqrt{1-w}\, B_{\mathrm{on}} + \sqrt{w}\, e^{i\varphi} B_{\mathrm{off}}$$

of a carrier-on component and a carrierless shared component. On this state pair, the diagonal readout R (bin-additive prediction, error $\leq 3.3 \times 10^{-16}$) and the coherent modulation $M = \sin 2\theta \cdot w\,|c|$ (error $\leq 6.1 \times 10^{-16}$) held **simultaneously**. The coexistence of the two grammars is not a switch between configurations but a property of a single state pair.

(iv) Distance law of the coherent coupling. The B wave packet was translated exactly along χ and the coupling $\langle a|b(d)\rangle$ was measured. The flow identity is exact at all separations ($\leq 1.4 \times 10^{-14}$), but the distance law is **entirely determined by the spectral envelope of the wave packet**: for equally spaced comb states the coupling does not decay but revives quasi-periodically (0.97 even at $d \geq 32$), while for Gaussian-envelope packets it decays by a factor of about 400 over the correlation length. In neither case does a $1/d^2$ law appear. **The original prediction of a "short-range type" was refuted** (correction record (v)). As a consequence, the Coulomb long-range law $1/r^2$ cannot arise from direct overlap; a mediating mechanism (harmonic closure [3]) must carry it — this experimentally pinpoints what the distance dictionary (Open Problem 2) must translate.

6.5 Exclusion of the Reversed Assignment — Uniqueness of the Two-Grammar Correspondence

With the measurements of Sections 3–6 in hand, the "one-to-one correspondence" of the property table can be fixed as a corollary via an exclusion argument. This section contains no new experiment — what follows is a logical consequence solely of rows already measured and committed.

Corollary (exclusion of the reversed assignment). In this system, the assignment of the magnitude term to the gravity-type grammar and of the overlap term to the charge-type grammar is not an arbitrary naming based on resemblance of properties but a unique correspondence that excludes the reversed assignment.

Proof. The magnitude term $\sin^2\theta\,(N_A - N_B)$ is a function of norms alone and is identically invariant under independent channel rephasings (Section 3, algebraic). Its composition law is exactly additive (mean, Experiment I, error $\leq 7.2 \times 10^{-16}$), and no product structure exists in the diagonal sector (zeroth-order no-go). The magnitude term therefore possesses neither the two signs via relative phase, nor flow reversal under phase reversal, nor an amplitude product, nor neutralization via channel non-sharing, and **cannot be the charge-type grammar**.

The overlap term $\sin 2\theta\,\mathrm{Re}\langle a|b\rangle$, on the other hand, reads the amplitude product on shared modes (Section 6.3, vertex-product law $\leq 4.0 \times 10^{-15}$), reverses sign under a half turn of the relative phase (measured in Section 6.3; exactly mirror-symmetric even over many collisions, Section 6.4 (ii), 2.2×10^{-16}), and vanishes for zero components or non-shared channels (neutrality $\leq 4.2 \times 10^{-15}$; null theorem, Section 6.1). Reading it as the gravity-type grammar would make the sign of the gravitational charge reverse under an internal phase flip and make gravitational action vanish under orthogonalization / decoherence, contradicting the requirements of the gravity-type grammar as a phase-blind, universal readout of magnitude (universality, no sign, unscreenable). **Hence the overlap term cannot be the gravity-type grammar.**

Since, by the classification theorem of Section 4, no third term exists in this class, no assignment other than

$$\boxed{\text{magnitude term} = \text{gravity-type grammar}, \qquad \text{overlap term} = \text{charge-type grammar}}$$

exists. ■

What this corollary fixes is the **pre-geometric assignment of interaction grammars**. The mapping to distance, direction, and metric on the three emergent directions, and the identification with the physical gravitational and Coulomb fields, remain as the separate tasks of the scale dictionary (Open Problem 2) and the translation of sign (Open Problem 3) — this limitation of scope is not meant to weaken the claim but to fix rigorously the domain of the uniqueness proposition.

7. Experiment III: The Quantization Mechanism (Rational Locking) — Summary of Sister Paper C

This section is the summary, required for this paper's claims (the quantization row of the property table), of the experiments reported in full in sister paper C (lock dynamics).

7.1 Zeroth-Order No-Go and the Working Hypothesis

By the theorem of Section 4, under a phase-blind readout θ is a constant of the motion (numerically confirmed: max drift 2.2×10^{-16} over 400 collisions $\times$ 4 amplitudes), and neither θ dynamics nor locking exists. Introducing a phase-sensitive readout (the power ratio of the symmetric channel $X = A + B$) as a conservative working hypothesis makes θ evolve.

7.2 Measurement of the Arnold Tongue

Sweeping the initial B amplitude, we found a finite interval where the tail average of the late-time rotation number pins exactly to a rational: for amplitudes 2.52–3.3,

$$\overline{\theta/\pi} = 1/3 \quad \text{(800 collisions, tail-300 average, } |\overline{\theta/\pi} - 1/3| \leq 5.6 \times 10^{-17}\text{)}$$

θ itself oscillates (tail standard deviation $\sim 7 \times 10^{-2}$), and only the mean rotation number locks to a rational — a periodic orbit exhibiting the standard behavior of an Arnold tongue. In the root convention $\theta = \pi/2 - \pi m/n$ this corresponds to $(m, n) = (1, 6)$, an order-6 lock. Separately, with the counting construction alone (zero amplitude search), an order-4 lock ($R = 1/2$ exact, period-8 recurrence, integer reconstruction of $(m, n) = (1, 4)$) was also obtained. The mechanism of quantization plateaus really exists in an endogenous-coupling system whose coupling is not supplied externally.

Note that the point at which the counting construction (equal-weight points = rational R) can connect directly to rational-angle locking (θ/π rational) is, by Niven's theorem (when θ/π is rational, $\cos\theta$ is rational only for $\cos\theta \in \{0, \pm\frac{1}{2}, \pm 1\}$ [17]), exactly limited to the five points

$$R \in \{0,\ 1/4,\ 1/2,\ 3/4,\ 1\}$$

Of these, this paper demonstrated $R = 1/2$ (period 8). Verification of the remaining nontrivial points $R = 1/4$ (predicted period 12) and $R = 3/4$ (predicted period 6) is carried by sister paper B (under the even≥2 variant of the convention audit, it has already been measured that the counting state lands exactly on $R = 3/4$).

7.3 Record of the Limits of the Search

Just below the lower edge of the tongue, the rotation number varies non-monotonically by $\sim 10^{-2}$ between adjacent amplitude points (a symptom of the supercritical regime), so a naive sweep for high-order tongues is unsuitable. The closest approach to the target $\rho = 39/124$ was 3.3×10^{-4} (1500 collisions, scan width 0.002) with no plateau detected — recorded as an upper bound on the tongue width. In addition, the hypothesis of obtaining the winding number $m = 23$ from a static state decomposition has been refuted (winding-number spectroscopy experiment: the state spectrum agrees completely with mapping artifacts; the $k = 23$ occupancy is at random level).

8. Convention Audit — Summary of Sister Paper B

This section is the summary, required for the convention robustness of this paper's claims, of the audit reported in full in sister paper B (structure of counting readouts). The definition of the fermion mask was varied over five variants (even≥4 / even≥6 / even≥2 / any≥4 / parity-reversed odd≥3) to audit the convention dependence of the claims.

Physics-grade (survives all conventions): that the readout of the equal-weight point is predicted by counting from the measured bin table $\times$ mask (20/20, $\leq 8 \times 10^{-16}$); the small-denominator rationality of the equal-weight points; the amplitude dependence; the counting-to-lock beachhead ($R = 1/2$, period 8, $(1, 4)$). In particular, under the parity-reversed mask the lock support moves to the 30 even harmonics — **the assignment "odd harmonics = fermionic" is a convention label, and the structure (counting $\rightarrow$ lock) is independent of the label**.

Bookkeeping-grade (products of the convention): the concrete form of the counting rule, the concrete values of the equal-weight points, the assignment of the odd/even roles, the existence condition of the beachhead. All claims of this paper are stated in physics-grade form.

9. Claims

Claim 1 (principal; discovery of the mechanism). The two-body flow of an anonymous closed wave system decomposes identically into two terms, and no third term exists. The two terms read exclusively the two pieces of information in the wave shape (norm / relative phase on shared modes), and interaction necessarily differentiates into exactly two grammars:

	Magnitude term	Overlap term
Composition law	**Additive** (mean, $\leq 7 \times 10^{-16}$)	**Product** $\sqrt{f_A f_B}$ ($\leq 4 \times 10^{-15}$)
Sign	None	Present (reverses with relative phase, measured)
Screening	Impossible (always on)	Possible (two mechanisms: zero component / non-sharing)
Quantization	None (continuous)	Present (rational locking, measured)
Range of the coupling	Ratio $(R/R_0)^2$: unbounded, continuous	Angle: $\alpha \leq 1/4\pi$ and lock discretization confine it to the $O(10^{-1})$ band

This property table corresponds one-to-one with the contrast table of gravity and electromagnetism. Moreover, this correspondence is **uniquely determined by an exclusion argument**, since both directions of the reversed assignment (overlap term → gravity-type / magnitude term → charge-type) contradict properties already measured (Section 6.5, Corollary). The differentiation of the forces requires no external input of particle species, coupling constants, or background.

Claim 2 (completeness on the Coulomb side). The overlap term possesses all five elements of the Coulomb structure — inverse square (derived in [3]), product, sign, quantization, and value (the exact finite-order root $\sin^2(23\pi/124)$, matching $\sqrt{4\pi\alpha}$ to 0.16 ppm [2]). We call this quantity **the model's elementary charge**.

Claim 3 (hierarchy). A long-range interaction that is weaker by orders of magnitude can exist only in the grammar of the magnitude term (the overlap term, squeezed between the unitarity bound and quantization, cannot leave the angular band). Moreover, the coupling of the magnitude term is of scale-ratio type $(R/R_0)^2$, and in any universe containing localized closures $R \ll R_0$; that is, **the enormous hierarchy between the two great long-range forces is a structure that follows from the definition of localization — particles are far smaller than the background**. The premise (gravitational coupling = norm ratio) and the derivation of the hierarchy's exponent (the scale dictionary) are left as explicit tasks.

Claim 4 (the one remaining item). Toward identification with the physical elementary charge, the only work remaining inside the model is the derivation of the mechanism that selects order 124 (the overall picture of the tasks connecting to emergent spacetime and physical forces is organized, as a series separate from the internal selection problem, in the closing section of sister paper C). The candidates have been narrowed by elimination through the experiments of sister paper C: the static decomposition does not select (refuted); the fixed-point-type channel shows no rational concentration (eliminated); the tongue weights of locking collapse with the denominator at a measured ratio > 461 (insufficient); and observational selection shows fidelity 1 at exact roots together with **inheritance of the selection into the divisor structure of the observation clock** (an observer with clock $J = 8$ cannot see the order-124 recurrence, while one with $J = 248$ can). The selection problem has therefore been transformed into the single falsifiable question **"why is the observation clock commensurable with 248?"**.

The sign-discrimination problem (prediction). The sign was measured as a relative phase. Its translation into spatial attraction/repulsion requires the distance dictionary ($\Delta\theta \leftrightarrow r$, unfinished), but the moment the dictionary is completed, whether this system is Bjerknes-type (equal phase = attraction) or electromagnetic-type (equal sign = repulsion) is uniquely determined by the structure of the dictionary — the sign problem is not an ambiguity but an automatically verified prediction.

10. Convergence with Known Systems

The force law product × cos(relative phase) × inverse square is the structure that Bjerknes (1875) demonstrated both theoretically and experimentally with pulsating spheres in an ideal fluid (alive in modern acoustics as the secondary Bjerknes force); that mode overlap is the existence condition of coupling is realized by visibility and distinguishability in optics (including Hong–Ou–Mandel); quantization plateaus via phase locking by the Shapiro steps of superconductivity (and, recently, current quantization via Bloch-oscillation synchronization); and the correspondence between the force ratio $\sim 10^{40}$ and scale ratios by Dirac's large numbers (1937) — each independently realized and observed. That fluids, optics, and superconductivity — mutually unrelated — and this model converge on the same grammar is evidence that this grammar is a general structure of closed wave systems rather than a detail of the medium.

The term "emergent electromagnetism" also appears in recent canonical-theoretic work (Phys. Rev. D **110**, 125006, 2024), but the distinction made there (the canonical structure of fundamental/emergent fields) and the distinction made here (the exclusive readout of the two terms of the flow) are different things. The contributions specific to this paper are: (i) the classification theorem for readouts and the composition laws in an endogenous-coupling system in which the coupling is read from the state rather than being an instrument constant; (ii) that the five elements and the two grammars hold simultaneously as terms of a single identity of a single system; (iii) the machine-precision verification of every row together with the convention audit.

11. Open Problems

1. **Selection problem**: identification of the endogenous dynamics that selects order 124. The candidates have been greatly narrowed by the experiments of sister paper C — the fixed-point-type channel is eliminated (the attractor set is a continuous band, with no rational concentration); by the measured tongue-width ratio (denominator 3 vs denominator ≥ 4, > 461) the lock weights are insufficient; observational selection shows fidelity 1 at exact roots and **inheritance of the selection into the divisor structure of the observation clock**, but finite-count observation can narrow only down to the recurrence basin. The current form of the question is therefore **"why is the observation clock commensurable with 248?"**, and this final form is taken up by the parent paper (the main paper 9)
2. **Scale dictionary**: the distance translation $\Delta\theta_n \leftrightarrow r$ and the value of R/R_0 (the exponent of the hierarchy). Section 6.4 (iv) identified what the dictionary must translate: since direct overlap does not produce $1/r^2$, the dictionary must translate **the distance readout of the mediating mechanism (harmonic closure)**
3. **Translation of sign**: discrimination between Bjerknes-type and electromagnetic-type (decided automatically once the dictionary is complete)
4. ~~Partial-sharing model~~ **Solved** (Section 6.4 (iii): demonstrated that the two grammars hold simultaneously on the same state pair)
5. **General form of the counting rule**: the closed form of the counting structure of equal-weight points and the general theory of integer-pair readout **are carried by sister paper B**

12. Reproducibility

All experiments referenced a single core runner (unchanged θ readout and rotation) with SHA-256 verification, were conducted by writing predicted values as in-code constants before measurement, and by placing reproductions of known results (anchors) at the head of each runner. Refuting experiments, invalid experiments, and corrections (including the correction of the R4 conservation misidentification) were all recorded without deletion. Experiment folders and commits:

Experiment	Folder	Commit
Two-memory discrimination battery	`two_memory_readout_battery_pre_v1`	5a44fbdb
Counting-rule verification + direct counting-lock bridge	`counting_rule_lock_bridge_pre_v1`	8dff131e
Winding-number spectroscopy (refutation)	`winding_spectroscopy_pre_v1`	641048ad
Convention audit	`convention_audit_pre_v1`	5a8f4b0a

Experiment	Folder	Commit
Mode-locking probe (Z_6 discovery)	`mode_locking_probe_pre_v1`	56453653
Readout classification + tongue spectroscopy	`tongue_spectroscopy_pre_v1`	913756fe
Mixed coupling (mean law)	`mixed_coupling_pre_v1`	fd7a3745
Cross-term charge grammar (v1 null / v2 invalid / v3 established)	`cross_term_charge_pre_v1`	26f26eb8
Batch regeneration of all figures in this paper	`paperA_two_grammar_evidence_v1`	381c2c5c
Niven intersection locks (periods 6/8/12, all predictions confirmed)	`niven_points_lock_pre_v1`	8f99d5c6
Claim hardening (multiple channels / sign persistence / mean-law convention extension)	`claim_hardening_battery_v1`	47e2da02
Partial sharing + fixed-point map	`partial_sharing_fixed_point_v1`	47e2da02
Distance law (comb/Gaussian contrast)	`coupling_range_v1`	47e2da02
Observation selection (clock selectivity)	`observation_selection_v1`	47e2da02
Tongue-width measurement	`tongue_width_measurement_v1`	47e2da02

Figures (all regenerable by the single runner of `paperA_two_grammar_evidence_v1`):

- Fig. A `figA_carrier_overlap_v1`: overlap matrices (with carrier = all zeros / without = ladder $\Delta k = \pm 2$)
- Fig. B `figB_classification_v1`: conserved/dynamical classification of the 7 readouts
- Fig. C `figC_mean_law_v1`: mean composition law and residuals
- Fig. D `figD_charge_grammar_v1`: sign reversal of $dN_B(\varphi)$ and the vertex-product line
- Fig. E `figE_three_terms_v1`: three-term decomposition, prediction vs. measurement (equal/unequal norm)

All measured-row CSVs were bundled by explicitly overriding the repository's exclusion rules.

References

Self-citations

1. Noriaki Kihara, "Anonymous Equal-Amplitude Composite-Wave Model: Basic Axiom System v6", Version DOI: `10.5281/zenodo.21465984`, Concept DOI: `10.5281/zenodo.21315735`, 2026.
2. Noriaki Kihara, "Discovery of Finite-Order Resonance in Iterated Exchange Scattering — Identifying the Origin of Peaks near Fine-Structure-Constant Values 137 and 128 with a Reproducible Wave-Packet Model", Version DOI: `10.5281/zenodo.21421367`, Concept DOI: `10.5281/zenodo.21421366`, 2026. (The two-channel exchange operator U_R and the eigenvalue structure $\lambda_a = e^{-2\pi i m/n}$, the B63 construction, and the defining source of the exact finite-order roots)
3. Noriaki Kihara, "Future Phase-Position Acceleration Map and the Inverse-Square Law via Harmonic Closure in an AB Two-Body Closed Phase System v4", Version DOI: `10.5281/zenodo.21468270`, Concept DOI: `10.5281/zenodo.21441081`, 2026. (Derivation of the inverse-square law, and the precedent for the three-step procedure zeroth-order no-go → conservative working hypothesis → independent verification)
4. Noriaki Kihara, "Where Are Mass, Momentum, and Energy Read From? — Multigauge Conserved Readouts of the ABC Closed Phase System", Version DOI: `10.5281/zenodo.21308050`, Concept DOI: `10.5281/zenodo.21308049`, 2026.

5. Noriaki Kihara, "Preliminary Summary of Acceleration-Basis and Localization Exchange in Exchange-Interference Scattering-Matrix Fermion-Like Collisions v1", Version DOI: `10.5281/zenodo.21333768`, Concept DOI: `10.5281/zenodo.21333766`, 2026. (Code origin of the scattering engine)
6. Noriaki Kihara, "Selection of the State-Exchange Weight G_R = 1 − R and Candidate Correspondence with the Fine-Structure Constant: A Numerical Experiment v1", Version DOI: `10.5281/zenodo.21396761`, Concept DOI: `10.5281/zenodo.21396760`, 2026. (Origin of the System A code)

- Sister paper B: Noriaki Kihara, "Structure of Counting Readouts in an Anonymous Two-Channel Closed Wave System" (Part II of the trilogy), Version DOI: `10.5281/zenodo.21763998`, Concept DOI: `10.5281/zenodo.21763997`, 2026.
- Sister paper C: Noriaki Kihara, "Lock Dynamics in an Anonymous Two-Channel Closed Wave System" (Part III of the trilogy), Version DOI: `10.5281/zenodo.21764000`, Concept DOI: `10.5281/zenodo.21763999`, 2026.

External References

The external references are not the basis of the derivations in this paper. They are used to show independent realizations of the same structure (evidence of convergence) and historical precedence.

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8. L. A. Crum, "Bjerknes forces on bubbles in a stationary sound field", *J. Acoust. Soc. Am.* **57**, 1363–1370 (1975). DOI: `10.1121/1.380614`. (The secondary Bjerknes force: amplitude product × cos(phase difference) × inverse square; in-phase attraction, antiphase repulsion)
9. M. Born and E. Wolf, *Principles of Optics*, 7th ed., Cambridge University Press (1999). (Two-beam interference law, mutual coherence)
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12. S. Shapiro, "Josephson currents in superconducting tunneling: the effect of microwaves and other observations", *Phys. Rev. Lett.* **11**, 80 (1963). DOI: `10.1103/PhysRevLett.11.80`. (Quantization plateaus via phase locking)
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14. P. A. M. Dirac, "The cosmological constants", *Nature* **139**, 323 (1937). DOI: `10.1038/139323a0`. (Large-number correspondence between force ratios and scale ratios. The same hypothesis's prediction of a time-varying G is not forced by the framework of this paper)
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16. E. I. Duque, "Emergent electromagnetism", *Phys. Rev. D* **110**, 125006 (2024). DOI: `10.1103/PhysRevD.110.125006`. arXiv:2407.14954. (The term is close, but it concerns the fundamental-field/emergent-field distinction via canonical structure, distinct from the two-term decomposition of this paper)
17. I. Niven, *Irrational Numbers*, Carus Mathematical Monographs No. 11, Mathematical Association of America (1956). (Theorem on the rationality of cosines of rational angles; the basis for the intersection set $\{0, 1/4, 1/2, 3/4, 1\}$ of the counting rational-R family and the rational-angle lock family)

Addendum (record of corrections) The following errors occurred and were corrected in the course of this work. (i) The cross-spectrum readout R4 was misidentified as a conserved readout (tail freezing misread as conservation) → the classification experiment revealed fixed-point convergence, and it was corrected. (ii) A mix-up in the sign prediction of the offset term of the flow equation → resolved by calibrating the rotation convention ($\sigma = +1$) (the exact agreement of the error $0.389 = \sin 0.4$ was the evidence of the sign slip). (iii) Collapse of the fraction-tuning assumption in the carrierless configuration (v2 invalid). (iv) The error in a prior record stating that the intersection of the counting-rational family and the rational-angle lock family is “$R = 1/2$ only” $\rightarrow$ corrected by Niven’s theorem to the five points $\{0, 1/4, 1/2, 3/4, 1\}$ (the correction was appended to the experiment folder README, and the remaining two points were confirmed with all predictions confirmed in the Niven intersection experiment). (v) The erroneous prediction that the distance law of the coherent coupling is “short-range type” $\rightarrow$ measurement revealed undamped revival for comb states and envelope decay for Gaussian packets; the distance law is determined by the packet envelope (Section 6.4 (iv)). All are recorded in the main text and in the reproduction package.